Type MS · Getting a bracket ready for the binomial series: a(1 + bx)^n

Answer 1 of 9 · Question 5 part a · 2 marks

2022 JUN · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

5 a	$(2+3x)^{-1}=\frac{1}{2}\left(1+\frac{3}{2}x\right)^{-1}$ so $p=\frac{1}{2}$ and $q=\frac{3}{2}$ oe	B1 B1 (2)
b	$\left(\frac{1}{2}\right)\left[1+(-1)\left("\frac{3}{2}"x\right)+\frac{(-1)(-2)}{2!}\left("\frac{3}{2}"x\right)^2+\frac{(-1)(-2)(-3)}{3!}\left("\frac{3}{2}"x\right)^3+\ldots\right]$	M1 A1ft
	$\frac{1}{2}-\frac{3}{4}x+\frac{9}{8}x^2-\frac{27}{16}x^3+\ldots$	A1 (3)
c	$\frac{1}{2}-\frac{3}{4}x+\frac{9}{8}x^2-\frac{27}{16}x^3+\frac{1}{2}x-\frac{3}{4}x^2+\frac{9}{8}x^3+\ldots$	M1
	$\frac{1}{2}-\frac{1}{4}x+\frac{3}{8}x^2-\frac{9}{16}x^3$	A1 (2)
d	$\int_0^{0.5}\left(\frac{1}{2}-\frac{1}{4}x+\frac{3}{8}x^2-\frac{9}{16}x^3\right)dx=\frac{1}{2}x-\frac{1}{8}x^2+\frac{1}{8}x^3-\frac{9}{64}x^4$	M1 A1
	$\frac{1}{2}\left(\frac{1}{2}\right)-\frac{1}{8}\left(\frac{1}{2}\right)^2+\frac{1}{8}\left(\frac{1}{2}\right)^3-\frac{9}{64}\left(\frac{1}{2}\right)^4=0.2256$ for awrt 0.2256	M1 A1 (4)
		Total 11 marks

Type MS · Getting a bracket ready for the binomial series: a(1 + bx)^n

Answer 1 of 9 · Question 5 part a · 2 marks

2022 JUN · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract

Part	Marks	Notes
(a)	B1	For $p=\frac{1}{2}$ or 2^{-1} or $q=\frac{3}{2}$
	B1	For $p=\frac{1}{2}$ or 2^{-1} and $q=\frac{3}{2}$
		NB $\frac{1}{2}\left(1+\frac{3}{2}x\right)^{-1}$ scores B1 B1
(b)	M1	For an attempt to expand $(1+qx)^{-1}$ with their value of q up to the term in x^3 It is not necessary to see p at this stage. The definition of an attempt is as follows: • The first term must be 1 • The next term must be correct for their value of q • The powers of qx must be correct eg $(qx)^2$ • The denominators must be correct Simplification not required. Do not allow missing brackets unless recovered later – this is a general point of marking.
	A1ft	For at least 3 terms fully correct and unsimplified for their value of q. It is not necessary to see p at this point.
	A1	For all 4 terms correct, all simplified.
	If there are any other methods used – send this to review please.	
(c)	M1	For multiplying their expansion by $(1+x)$ There must be a clear attempt to multiply to get 7 terms, allow up to 2 errors. Ignore terms with powers higher than 3.
	A1	For all 4 terms correct, ignore terms with powers higher than 3.
(d)	M1	For a minimally acceptable attempt to integrate an expression with at least 3 terms, which must include a term in x^3 See general guidance for definition of minimally acceptable attempt.
	A1ft	For a fully correct integration of their expression (defined as above for the M mark).
	M1	For substitution of limits into a changed expression, the correct way round – this can be implied if the final answer is correct. We don't need to see 0 substituted in.
	A1	For awrt 0.2256 (Note the calculator value is 0.2288461……)

Type MS · Getting a bracket ready for the binomial series: a(1 + bx)^n

Answer 1 of 9 · Question 5 part a · 2 marks

2022 JUN · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme								Marks
	0	*0.25*	*0.5*	**1**	*1.5*	**2**	3		B2

Type MS · Getting a bracket ready for the binomial series: a(1 + bx)^n

Answer 2 of 9 · Question 6 part a · 2 marks

2023 JAN · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

6 a	$p=2 \quad q=\frac{3}{8}$	B1 B1 (2)
b	$(2)\left[1+\frac{1}{3}\left("\frac{3}{8}"x\right)+\frac{\frac{1}{3}\left(-\frac{2}{3}\right)}{2!}\left("\frac{3}{8}"x\right)^2\right]=2+\frac{1}{4}x-\frac{1}{32}x^2$	M1 A1ft A1 (3)
c	$\left(\sqrt[3]{9}=\sqrt[3]{8+1}\Rightarrow x=\frac{1}{3}\right)$ So $"2"+"\frac{1}{4}"\times"\frac{1}{3}"-"\frac{1}{32}"\times\left("\frac{1}{3}"\right)^2\left(=\frac{576+24-1}{288}\right)=\frac{599}{288}$ *	M1 A1 cso (2)
		Total 7 marks

Part	Mark	Additional Guidance
(a)	B1	For $p=2$ **or** $q=\frac{3}{8}$ condone $8^{\frac{1}{3}}$
	B1	For $p=2$ **and** $q=\frac{3}{8}$ condone $8^{\frac{1}{3}}$
(b)	M1	For an attempt to expand $(1+qx)^{\frac{1}{3}}$ with their value of q up to the term in x^2 It is not necessary to see p at this stage. The definition of an attempt is as follows: • The first term must be 1 • The next term must be correct for their value of q • The powers of qx must be correct eg $(qx)^2$ • The denominators must be correct Simplification not required. Do not allow missing brackets unless recovered later – this is a general point of marking. Ignore any terms with powers higher than 2.
	A1ft	For both algebraic terms fully correct and unsimplified in the expansion of $(1+qx)^{\frac{1}{3}}$ for their value of q. It is not necessary to see p at this point. Ignore any terms with powers higher than 2.
	A1	For all 3 terms correct, all simplified. Ignore any terms with powers higher than 2
If there are any other methods used – send this to review please.		
(c)	M1	For correct substitution of $x=\frac{1}{3}$ into their expansion, which must have at least 2 terms, including a term in x^2
	A1cso*	Obtains the given result

Type MS · Getting a bracket ready for the binomial series: a(1 + bx)^n

Answer 2 of 9 · Question 6 part a · 2 marks

2023 JAN · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Scheme	Marks
	-6 -5 -4 –3 –2 –1 0	B2

Type MS · Getting a bracket ready for the binomial series: a(1 + bx)^n

Answer 3 of 9 · Question 6 part a · 2 marks

2025 JUN · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

6 (a)	$\sqrt{4-x}=2\left(1-\frac{x}{4}\right)^{\frac{1}{2}}$ or $A=2, B=4$	B1B1 [2]
(b)	$2\left\{1+\left(\frac{1}{2}\right)\left(-\frac{x}{4}\right)+\frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(-\frac{x}{4}\right)^2}{2!}+\frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{x}{4}\right)^3}{3!}+...\right\}$ $=2-\frac{x}{4}-\frac{x^2}{64}-\frac{x^3}{512}+...$	M1A1ft A1 [3]
(c)	$\frac{\sqrt{305}}{10}=\sqrt{4-x}\Rightarrow x=0.95$ $2-\frac{0.95}{4}-\frac{0.95^2}{64}-\frac{0.95^3}{512}=1.7467238.....\approx 1.74672$	B1 M1A1 [3]
		Total 8 marks

Part	Mark	Additional Guidance
(a)	B1B1	B1B0 for one correct value, B1B1 For the correct values of both A and B A and B must be integers from correct working **Do not accept** $A=4^{\frac{1}{2}}$ or $\sqrt{4}$
(b)	M1	Attempt to expand their $'2'\left(1-\frac{x}{'4'}\right)^{\frac{1}{2}}$ up to x^3 term (with or without '2') Must have: • First term is 1 for their $\left(1-\frac{x}{'4'}\right)^{\frac{1}{2}}$ expansion • Correct powers of $\pm\frac{x}{'4'}$ for each term No simplification needed. Ignore terms beyond x^3
	A1ft	Two correct algebraic terms in their expansion, with or without'2' Not necessarily in lowest terms. Ignore terms beyond x^3.
	A1	All terms correct and in lowest terms. Ignore terms beyond x^3.
(c)	B1	Identifies $x=0.95$ oe. e.g. $\frac{19}{20}$
	M1	Substitutes their x which must come from some working (provided $\lvert x\rvert<4$) into each of the algebraic terms of their expansion.
	A1	1.74672 (must be 5dp) **Note:** Calculator value for $\frac{\sqrt{305}}{10}$ is 1.74642492

Type MS · Getting a bracket ready for the binomial series: a(1 + bx)^n

Answer 3 of 9 · Question 6 part a · 2 marks

2025 JUN · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Getting a bracket ready for the binomial series: a(1 + bx)^n

Answer 4 of 9 · Question 9 part a · 2 marks

2020 JAN · Paper 02 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

9 (a)	$\frac{1}{(2-x)^3} = (2-x)^{-3} = \frac{1}{8}\left(1-\frac{x}{2}\right)^{-3} \Rightarrow p = \frac{1}{8},\ q = \frac{1}{2}$	B1B1 [2]
(b)	$\frac{1}{8}\left(1-\frac{x}{2}\right)^{-3} = \frac{1}{8}\left[1+(-3)\left(-\frac{x}{2}\right)+\frac{(-3)(-4)\left(-\frac{x}{2}\right)^2}{2!}+\frac{(-3)(-4)(-5)\left(-\frac{x}{2}\right)^3}{3!}\right]$ $= \frac{1}{8}+\frac{3}{16}x+\frac{3}{16}x^2+\frac{5}{32}x^3+\ldots$	M1 A1A1 [3]
(c)	$(a+bx)\left(\frac{1}{8}+\frac{3}{16}x+\frac{3}{16}x^2\right) = \frac{a}{8}+x\left(\frac{2b}{16}+\frac{3a}{16}\right)+\left\{x^2\left(\frac{3a}{16}+\frac{3b}{16}\right)\right\}$ $\Rightarrow \frac{3}{8} = \frac{a}{8} \Rightarrow a = 3$ $\Rightarrow -\frac{43}{16} = \frac{2b+3a}{16} \Rightarrow 2b = -43-9 = -52 \Rightarrow b = -26$	M1 A1 A1 [3]
(d)	$\frac{3a+3b}{16} = \frac{9-78}{16} = -\frac{69}{16}$ oe	M1A1 [2]
		Total 10 marks

Type MS · Getting a bracket ready for the binomial series: a(1 + bx)^n

Answer 4 of 9 · Question 9 part a · 2 marks

2020 JAN · Paper 02 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

(b)	
M1	Attempts to use the binomial expansion for their $(1-qx)^{-3}$. Must have first term 1, three more terms with ascending powers of x, 2 or 2! and 6 or 3! seen , and their $\left(-\frac{x}{2}\right)$ used at least once. No simplification needed. Ignore terms beyond x^3
A1	Two algebraic terms correct in the expansion for their $(1-qx)^{-3}$.. Must be single fractions, not necessarily in lowest terms. Ignore terms beyond x^3
A1	All four terms correct and in lowest terms. Ignore terms beyond x^3
(c)	
M1	For either their $\frac{1}{8}a=\frac{3}{8}$ or their $\frac{3}{16}a+$ their $\frac{1}{8}b=-\frac{43}{16}$ May be implied by a correct value of a or b.
A1	$a=3$
A1	$b=-26$ **NB** answers $a=3$ and $b=-26$ scores 3/3
(d)	
M1	Substituting their a and their b into their $\frac{3a+3b}{16}$
A1	$-\frac{69}{16}$ oe **NB** If $p=2,\ q=\frac{1}{2}$ is used then $a=\frac{3}{16}$ and $b=-\frac{13}{8}$ which substituted into $3a+3b$ gives an answer of $-\frac{69}{16}$ but scores A0

Type MS · Getting a bracket ready for the binomial series: a(1 + bx)^n

Answer 4 of 9 · Question 9 part a · 2 marks

2020 JAN · Paper 02 (regional R) · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Getting a bracket ready for the binomial series: a(1 + bx)^n

Answer 5 of 9 · Question 9 part a · 2 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

		For attempt to integrate one of: $9-9e^{-3x}$ or $e^{3x}-1$ or $\pm[(9-9e^{-3x})-(e^{3x}-1)]$ Limits may not be present. At least one term correct. Ignore +c if included.
	A1$_2$	For correct integration of one of the exponential terms $\pm 9e^{-3x} \rightarrow \mp\frac{9}{3}e^{-3x}$ **or** $\pm e^{3x} \rightarrow \pm\frac{1}{3}e^{3x}$ Limits need not be present. Ignore +c if included.
	A1$_3$	For correct integration of both curves $9x+3e^{-3x}$ **and** $\frac{e^{3x}}{3}-x$ or for a **fully correct** integration where the difference between two expressions is found $\pm\left(10x-\frac{9}{-3}e^{-3x}-\frac{e^{3x}}{3}\right)$ or $\pm\left(9x-\frac{9}{-3}e^{-3x}-\frac{e^{3x}}{3}+x\right)$ **Note: this is an M mark in epen**
	M1$_4$	For the difference between the two expressions either before or after integration. Allow subtraction either way around. **Note: this is an A mark in epen**
	dM1$_5$	Substitution of correct limits into their integrated expressions (limits subtracted the correct way around). Dependent on first M scored. If substituting before difference found then must substitute into both integrated expressions. May be implied by awrt 1.99 If integration is not correct then substitution must be shown.
	A1$_6$	For the correct answer oe. Must be exact value.

Type MS · Getting a bracket ready for the binomial series: a(1 + bx)^n

Answer 5 of 9 · Question 9 part a · 2 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks
9(a)	$(3(3-x)^{-3} = \quad \frac{1}{9}\left(1-\frac{x}{3}\right)^{-3})$ $\quad a=\frac{1}{9} \quad b=\frac{1}{3}$	B1 B1 [2]
(b)	$\left(1-\frac{x}{3}\right)^{-3} =$	
	$\left[1+(-3)\left(-\frac{x}{3}\right)+\frac{(-3)(-4)\left(-\frac{x}{3}\right)^2}{2!}+\frac{(-3)(-4)(-5)\left(-\frac{x}{3}\right)^3}{3!}\right]$	M1 A1ft
	$\frac{1}{9}+\frac{1}{9}x+\frac{2}{27}x^2+\frac{10}{243}x^3$	A1 [3]
(c) (i)	$\frac{24}{125}=\frac{3}{(3-x)^3}$ or $\frac{125}{8}=(3-x)^3 \Rightarrow \frac{5}{2}=3-x$	
	$x=0.5$ oe	B1
	$\frac{1}{9}+\frac{1}{9}("0.5")+\frac{2}{27}("0.5")^2+\frac{10}{243}("0.5")^3 \quad (=0.19033)$	B1ft
(ii)	$\pm\left(\frac{\frac{24}{125}-"0.19033"}{\frac{24}{125}}[\times 100]\right)$ oe	M1
	0.87% or −0.87%	A1 [4]
		Total 9 marks

Type MS · Getting a bracket ready for the binomial series: a(1 + bx)^n

Answer 5 of 9 · Question 9 part a · 2 marks

2022 JUN · Paper 02 (regional R) · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Getting a bracket ready for the binomial series: a(1 + bx)^n

Answer 6 of 9 · Question 6 part a · 2 marks

2024 NOV · Paper 01 · find part a in this mark scheme extract

Number		
6(a)	eg $\frac{a}{\sqrt{4\left(1+\frac{bx}{4}\right)}}$ or $\frac{a}{\sqrt{4}\sqrt{\left(1+\frac{bx}{4}\right)}}$ or $\frac{a}{4^{\frac{1}{2}}\left(1+\frac{bx}{4}\right)^{\frac{1}{2}}}$ or $a\left(4^{-\frac{1}{2}}\right)\left(1+\frac{bx}{4}\right)^{-\frac{1}{2}}$ or $a\left(4\left(1+\frac{bx}{4}\right)\right)^{-\frac{1}{2}}$	M1
	$=\frac{a}{2}\left(1+\frac{bx}{4}\right)^{-\frac{1}{2}}$	A1*cso [2]
(b)	$\left(\frac{a}{2}\right)\left[1+\left(-\frac{1}{2}\right)\left(\frac{bx}{4}\right)+\frac{\left(-\frac{1}{2}\right)\left(-\frac{1}{2}-1\right)}{2!}\left(\frac{bx}{4}\right)^2+\frac{\left(-\frac{1}{2}\right)\left(-\frac{1}{2}-1\right)\left(-\frac{1}{2}-2\right)}{3!}\left(\frac{bx}{4}\right)^3\right]$ oe	M1 M1 A1
	$\left(\left(\frac{a}{2}\right)\left[1+\left(-\frac{bx}{8}\right)+\frac{3}{128}b^2x^2+\left(-\frac{5}{1024}b^3x^3\right)\right]\right)$ Not necessary - but must be checked if seen All intermediate steps must be checked for errors $\frac{a}{2}-\frac{ab}{16}x+\frac{3ab^2}{256}x^2-\frac{5ab^3}{2048}x^3 \quad P=\frac{a}{2}, Q=-\frac{ab}{16}*, R=\frac{3ab^2}{256}, S=-\frac{5ab^3}{2048}*$	A1*cso [4]
(c)	$-\frac{ab}{16}=\frac{128}{5}.\left(-\frac{5ab^3}{2048}\right)$ or $\frac{5}{128}.\left(-\frac{ab}{16}\right)=-\frac{5ab^3}{2048}$ or $-\frac{5ab}{2048}=-\frac{5ab^3}{2048}$ or $-\frac{ab}{16}=-\frac{ab^3}{16}$	M1
	$\left(b^2=1\right)b=1*$	A1*
	$\frac{3a\times1^2}{256}=\frac{9}{256}\Rightarrow a=3*$	B1*cso [3]
(d)	$\frac{\sqrt{6}}{2}=\frac{3}{2}\left(1+\frac{x}{4}\right)^{-\frac{1}{2}}$ or $\frac{\sqrt{6}}{2}=\frac{3}{\sqrt{4+x}}$ or $\frac{3}{\sqrt{6}}=\frac{3}{\sqrt{4+x}}$ or $\sqrt{6}=\sqrt{4+x}$ oe $\Rightarrow x=(2)$	M1
	$\frac{3}{2}-\frac{3\times1}{16}\times2+\frac{3\times3\times1^{(2)}}{256}\times2^2-\frac{5\times3\times1^{(3)}}{2048}\times2^3$	dM1
	1.207	A1 [3]
		Total 12 marks

Type MS · Getting a bracket ready for the binomial series: a(1 + bx)^n

Answer 7 of 9 · Question 6 part a · 2 marks

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6 (a)	$\sqrt{9-x}=3\left(1-\frac{x}{9}\right)^{\frac{1}{2}} \Rightarrow p=3, \quad q=-\frac{1}{9}$	B1B1 [2]
(b)	$3\left(1-\frac{x}{9}\right)^{\frac{1}{2}}=$ $3\left\{1+\left(\frac{1}{2}\right)\left(-\frac{x}{9}\right)+\frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(-\frac{x}{9}\right)^2}{2!}+\frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{x}{9}\right)^3}{3!}+\ldots\right\}$ $=3\left(1-\frac{x}{18}-\frac{x^2}{648}-\frac{x^3}{11664}\right)=3-\frac{x}{6}-\frac{x^2}{216}-\frac{x^3}{3888}$	M1 A1A1 [3]
(c)	$\frac{31}{4}=9-x \Rightarrow x=\frac{5}{4}$ $\sqrt{\frac{31}{4}} \approx 3-\frac{1.25}{6}-\frac{1.25^2}{216}-\frac{1.25^3}{3888}=2.783930523\ldots \approx 2.78393$	B1 M1A1 [3]
		Total 8 marks

(a)	B1	$p=3$ or $\sqrt{9}$ or $9^{\frac{1}{2}}$	May be shown in their $3\left(1-\frac{x}{9}\right)^{\frac{1}{2}}$ (allow isw)
	B1	$q=-\frac{1}{9}$	
(b)	M1	Attempt to use the binomial expansion for their $(1+qx)^{\frac{1}{2}}$. Must have first term 1, three more terms with ascending powers of x, 2 or 2! and 6 or 3! Seen, and their $\left(-\frac{x}{9}\right)$ used at least once. No simplification needed. Ignore terms beyond x^3	
	A1	Two algebraic terms correct in the expansion of $3\left(1-\frac{x}{9}\right)^{\frac{1}{2}}$. Must be single fractions, not necessarily in lowest terms. Ignore terms beyond x^3	
	A1	All four terms correct and in lowest terms. Ignore terms beyond x^3	
(c)	B1	Identify $x=\frac{5}{4}$ as the value needed to make $\sqrt{9-x}=\sqrt{\frac{31}{4}}$	
	M1	Substitute their x into each of the algebraic terms of their expansion of $\sqrt{9-x}$	
	A1	2.78393 **given to 5DP**.	

Type MS · Getting a bracket ready for the binomial series: a(1 + bx)^n

Answer 7 of 9 · Question 6 part a · 2 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Getting a bracket ready for the binomial series: a(1 + bx)^n

Answer 8 of 9 · Question 10 part a · 2 marks

2021 NOV · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

10 (a)	$\frac{3}{\sqrt{9-3x}}=\frac{1}{\sqrt{9}}\times\frac{3}{\sqrt{1-\frac{3x}{9}}}=\frac{1}{\sqrt{1-\frac{x}{3}}}=\left(1-\frac{x}{3}\right)^{-\frac{1}{2}}$ *	B1B1* cso [2]
(b)	$\left(1-\frac{x}{3}\right)^{-\frac{1}{2}}=1+\left(-\frac{1}{2}\right)\left(-\frac{x}{3}\right)+\frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{x}{3}\right)^2}{2!}+\frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{5}{2}\right)\left(-\frac{x}{3}\right)^3}{3!}$ $\left(1-\frac{x}{3}\right)^{-\frac{1}{2}}=1+\frac{x}{6}+\frac{x^2}{24}+\frac{5x^3}{432}$	M1 A1A1 [3]
(c)	$3\mathrm{f}(x)=\frac{3(1+2x)}{\sqrt{9-3x}}=\frac{3(1+2x)}{3\sqrt{1-\frac{x}{3}}}=(1+2x)\left(1-\frac{x}{3}\right)^{-\frac{1}{2}}$ $3\mathrm{f}(x)=(1+2x)\left(1+\frac{x}{6}+\frac{x^2}{24}+\frac{5x^3}{432}\right)$ $3\mathrm{f}(x)=(1+2x)\left(1+\frac{x}{6}+\frac{x^2}{24}+\frac{5x^3}{432}\right)=1+\frac{13}{6}x+\frac{3}{8}x^2+\frac{41}{432}x^3$	B1 M1 M1A1 [4]
(d)	$\int_{0.1}^{0.2}\frac{3+6x}{\sqrt{9-3x}}\,\mathrm{d}x=\int_{0.1}^{0.2}\left(1+\frac{13}{6}x+\frac{3}{8}x^2+\frac{41}{432}x^3\right)\mathrm{d}x$ $=\left[x+\frac{13}{12}x^2+\frac{3}{24}x^3+\frac{41}{1728}x^4\right]_{0.1}^{0.2}$ $\int_{0.1}^{0.2}\frac{1+2x}{\sqrt{9-3x}}\,\mathrm{d}x=\frac{1}{3}\left[\left(0.2+\frac{13}{12}\times0.2^2+\frac{3}{24}\times0.2^3+\frac{41}{1728}\times0.2^4\right)-\left(0.1+\frac{13}{12}\times0.1^2+\frac{3}{24}\times0.1^3+\frac{41}{1728}\times0.1^4\right)\right]=0.044\ 470\ 2$	M1A1ft dM1A1 [4]
		Total 13 marks

Type MS · Getting a bracket ready for the binomial series: a(1 + bx)^n

Answer 8 of 9 · Question 10 part a · 2 marks

2021 NOV · Paper 02 · extract 2 of 3 · find part a in this mark scheme extract

	B1*	For simplifying to the required form $\frac{3}{\sqrt{9-3x}}=\left(1-\frac{x}{3}\right)^{-\frac{1}{2}}$
(b)	**M1**	For an attempt to find the binomial expansion for the given expression. • The expansion must begin with 1 • The denominators must be correct (ie. 2! And 3!) on the third and fourth terms. • The power of x must be correct (ie. $\left(-\frac{x}{3}\right)$ $\left(-\frac{x}{3}\right)^2$ and $\left(-\frac{x}{3}\right)^3$ with the correct corresponding denominators.
	A1	For one algebraic term correct and simplified
	A1	For the whole expansion correct up to x^3 Ignore extra terms beyond these. $\left(1-\frac{x}{3}\right)^{-\frac{1}{2}}=1+\frac{x}{6}+\frac{x^2}{24}+\frac{5x^3}{432}$
(c)	**B1**	For writing down $3f(x)=(1+2x)\left(1-\frac{x}{3}\right)^{-\frac{1}{2}}$
	M1	For replacing $\left(1-\frac{x}{3}\right)^{-\frac{1}{2}}$ with their $\left(1+\frac{x}{6}+\frac{x^2}{24}+\frac{5x^3}{432}\right)$ provided their expansion has at least two terms one of which is an algebraic term.
	M1	For an attempt to expand $(1+2x)\left('1+\frac{x}{6}+\frac{x^2}{24}+\frac{5x^3}{432}'\right)$ up to and including the term in x^3. Accept the expansion of $(3+6x)\left('1+\frac{x}{6}+\frac{x^2}{24}+\frac{5x^3}{432}'\right)$
	A1	For the correct expansion $3f(x)=1+\frac{13}{6}x+\frac{3}{8}x^2+\frac{41}{432}x^3$ ignore extra terms
(d)	**M1**	For an attempt to integrate their expansion of 3f(x) or f(x) which must have as a minimum two terms, one of which is a term in x, x^2 or x^3 Ignore limits for this mark. They may multiply by $\frac{1}{3}$ at this stage so accept e.g $\left[\frac{x}{3}+\frac{13}{36}x^2+\frac{3}{72}x^3+\frac{41}{5184}x^4\right]$
	A1ft	For a correct integrated expression ft their expression.
	dM1	For substituting both values of 0.2 and 0.1 the correct way around seen. This mark can be implied by the value of 0.044 470 2 seen. If their expansion from (c) has not been divided by 3 for the integration, it must be divided by 3 here for the award of this mark.
	A1	For the correct value of 0.044 470 2 **NB**: The calculator value is 0.044 470 77

Type MS · Getting a bracket ready for the binomial series: a(1 + bx)^n

Answer 8 of 9 · Question 10 part a · 2 marks

2021 NOV · Paper 02 · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Getting a bracket ready for the binomial series: a(1 + bx)^n

Answer 9 of 9 · Question 7 part c · 2 marks

2023 JUN · Paper 02 · extract 1 of 3 · find part c in this mark scheme extract

7(a)	$\left(1+\frac{x}{3}\right)^{-3}=\left[1+(-3)\left(\frac{x}{3}\right)+\frac{(-3)(-3-1)}{2!}\left(\frac{x}{3}\right)^2+\frac{(-3)(-3-1)(-3-2)}{3!}\left(\frac{x}{3}\right)^3\ldots\right]$ $=1-x+\frac{2}{3}x^2-\frac{10}{27}x^3$	M1A1 A1 [3]
(b)	$-3<x<3$ or $\lvert x\rvert<3$	B1 [1]
(c)	$\left((3+x)^{-3}=3^{-3}\times\left(1+\frac{x}{3}\right)^{-3}=\frac{1}{27}\times\left(1+\frac{x}{3}\right)^{-3}\right)$ $P=\frac{1}{27}$, $Q=\frac{1}{3}$	 B1, B1 [2]
(d)	$\frac{(1+4x)}{(3+x)^3}=(1+4x)\times\frac{1}{27}\times\left(1-x+\frac{2}{3}x^2\right)$ $=\frac{1}{27}\times\left(1-x+\frac{2}{3}x^2+4x-4x^2+\ldots.\right)$ $=\frac{1}{27}\left(1+3x-\frac{10x^2}{3}\right)$ or $\frac{1}{27}+\frac{x}{9}-\frac{10x^2}{81}$	 M1 A1 [2]
(e)	$\int_0^{0.2}\left(\frac{1}{27}+\frac{x}{9}-\frac{10x^2}{81}\right)\mathrm{d}x=\left[\frac{x}{27}+\frac{x^2}{18}-\frac{10x^3}{243}\right]_0^{0.2}$ $=\left(\frac{0.2}{27}+\frac{0.2^2}{18}-\frac{10\times0.2^3}{243}\right)-[0]$ $=0.0093004$	M1 M1 A1 [3]
		Total 11 marks

Type MS · Getting a bracket ready for the binomial series: a(1 + bx)^n

Answer 9 of 9 · Question 7 part c · 2 marks

2023 JUN · Paper 02 · extract 2 of 3 · find part c in this mark scheme extract

Part	Mark	Notes
(a)	M1	For applying a correct binomial expansion in unsimplified form. Minimum required: • The expansion begins with 1 • The next term is correct • The powers of $\frac{x}{3}$ must be correct eg $\left(\frac{x}{3}\right)^2$ • The denominators are correct. Do not allow missing brackets unless recovered later – this is a general point of marking. Ignore any terms with powers higher than 3.
	A1	Following M1 (this is a general point of marking, A marks can only follow M marks), all conditions above met, must see $1 - x$ and at least the term in x^2 or x^3 correct and simplified. Ignore any terms with powers higher than 3.
	A1	A fully correct and simplified expansion. Ignore any terms with powers higher than 3..
(b)	B1	For the correct validity.
(c)	B1	For the correct value of P or Q explicitly written or embedded in $\frac{1}{27}\times\left(1+\frac{x}{3}\right)^{-3}$
	B1	For the correct values of P and Q explicitly written or embedded in $\frac{1}{27}\times\left(1+\frac{x}{3}\right)^{-3}$
(d)	M1	For attempting to multiply their expansion, which must be of the form "P"× (their expansion from part (a), with a minimum of 3 terms, by $(1 + 4x)$. An attempt must include 3 correctly multiplied out terms of their expansion before simplification. The "P" may remain factorised. Ignore any terms which to powers of x higher than 2.
	A1	For the correct expansion in either form shown. Allow equivalent coefficients. Ignore powers of x higher than 2.
(e)	M1	For an attempt to integrate their expression from part (d), provided it has at least one constant term and at least one algebraic term. See general guidance, no power of x must decrease.
	M1	For substituting in the value of 0.2 into a changed expression and subtracting the correct way. Must see the explicit substitution of 0.2 at least once, if the final answer is not correct. Can be implied if final answer correct. Substitution of 0 does not need to be seen.
	A1	For the value of 0.0093004 [The calculator value is 0.0093316] Accept the value of $\frac{113}{12150}$ if seen.

Type MS · Getting a bracket ready for the binomial series: a(1 + bx)^n

Answer 9 of 9 · Question 7 part c · 2 marks

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Question	Scheme	Marks